

On a Proof from the Calculus of Probability Due to Lejeune Dirichlet

After a communication by J. F. Encke

If the errors of observation are arranged according to their absolute magnitudes, without regard to their signs, then for m observations the error belonging to the index $\frac{1}{2}(m+1)$, when m is odd, or, when m is even, the arithmetic mean of the errors with indices $\frac{1}{2}m$ and $\frac{1}{2}m+1$, will give an approximate value for the probable error r .

If, already for power sums, the larger number of observational errors makes the accuracy with respect to probable limits increase so much, then with this method of counting this must happen all the more. Since GAUSS, in the *Zeitschrift für Astronomie*, vol. I, p. 195, gave the formula needed for this without proof, the following elegant proof, which I owe to the communication of my esteemed colleague Professor DIRICHLET, will be all the more valuable here, since the theorem itself has not yet been proved elsewhere.

We seek the probability that, among $(2n+1)$ observations, the distribution of errors is such that one error lies between t and $t+dt$, n errors lie between 0 and t , and n errors lie between $t+dt$ and ∞ . Let the probability that an error is smaller than t be, quite generally,

$$\int_0^t \psi(\Delta) d\Delta = u.$$

The probability of an error $> t+dt$ then becomes

$$1 - \psi(t)dt - \int_0^t \psi(\Delta) d\Delta = 1 - u - \psi(t)dt,$$

since the probability of an error between t and $t+dt$ is $\psi(t)dt$.

Thus the compound probability of one arrangement of the errors, with n errors $< t$, one error between t and $t+dt$, and n errors $> t+dt$, is

$$u^n(1-u)^n\psi(t)dt,$$

if terms of the second order are neglected, since the result is of the first order. But such cases or arrangements can occur in as many ways as there are permutations of $2n+1$ elements when among them there are n equal elements of one kind, with probability u , and n equal elements of another kind, with probability $1-u$. Consequently the probability of all possible arrangements of this kind is

$$\frac{1 \cdot 2 \cdot 3 \cdots (2n+1)}{(1 \cdot 2 \cdot 3 \cdots n)^2} u^n(1-u)^n\psi(t)dt = U.$$

If the size dt of the interval between t and $t+dt$ is held constant, there is a value of t for which U is a maximum. Differentiation gives, for its determination,

$$\frac{n\psi(t)}{u} - \frac{n\psi(t)}{1-u} + \psi'(t) = 0.$$

Here $\psi'(t)$ has the same meaning as $\psi'(\Delta)$ above; namely du , or the increment of $\int_0^t \psi(\Delta) d\Delta$ under an infinitely small change in the upper limit t , is equal to $\psi(t)dt$. The last equation can be put in the form

$$\frac{1}{u} - \frac{1}{1-u} + \frac{\psi'(t)}{n\psi(t)} = 0.$$

The last term becomes smaller the larger n is, or the more observations are given. For a sufficiently large number it may be neglected. Thus the value of t for which the maximum occurs approaches, as n increases, the value that follows from

$$\frac{1}{u} - \frac{1}{1-u} = 0,$$

that is,

$$u = \int_0^t \psi(\Delta) d\Delta = \frac{1}{2},$$

or, by the definition given above, the value r .

If one takes the integral of U between assigned limits, one obtains the probability that the middle error is contained within those limits. For the limits $r - \delta$ and $r + \delta$ this probability is therefore

$$\frac{1 \cdot 2 \cdot 3 \cdots (2n+1)}{(1 \cdot 2 \cdot 3 \cdots n)^2} \int_{r-\delta}^{r+\delta} u^n (1-u)^n \psi(t) dt.$$

Since $\psi(t)dt = du$, if with respect to t the limits are written

$$\int_0^{r-\delta} \psi(t)dt = u', \quad \int_0^{r+\delta} \psi(t)dt = u'',$$

the probability that the middle value lies between $r - \delta$ and $r + \delta$ becomes

$$\frac{1 \cdot 2 \cdot 3 \cdots (2n+1)}{(1 \cdot 2 \cdot 3 \cdots n)^2} \int_{u'}^{u''} u^n (1-u)^n du.$$

The larger the number of observations, the narrower the limits between which t will lie with equal probability. If the observations are sufficiently numerous, then in expanding u' and u'' by Taylor's theorem it is permissible to retain only the first power of δ . Hence

$$u' = \int_0^r \psi(t)dt - \delta\psi(r) \cdots = \frac{1}{2} - \delta\psi(r),$$

and similarly

$$u'' = \frac{1}{2} + \delta\psi(r).$$

Both this form and the combination of u and $1-u$ in the integral show that a more convenient form is obtained by introducing another variable for u , most simply by

$$u = \frac{1}{2} + \frac{s}{2\sqrt{n}} = \frac{1}{2} \left(1 + \frac{s}{\sqrt{n}} \right),$$

so that

$$1-u = \frac{1}{2} - \frac{s}{2\sqrt{n}} = \frac{1}{2} \left(1 - \frac{s}{\sqrt{n}} \right).$$

The corresponding limits in s are found from

$$\delta\psi(r) = \frac{s}{2\sqrt{n}}.$$

The integral therefore becomes

$$\frac{1 \cdot 2 \cdot 3 \cdots (2n+1)}{(1 \cdot 2 \cdot 3 \cdots n)^2} \cdot \frac{1}{2^{2n+1}\sqrt{n}} \int_{-2\delta\sqrt{n}\psi(r)}^{+2\delta\sqrt{n}\psi(r)} \left(1 - \frac{s^2}{n} \right)^n ds,$$

or, since only even powers of s occur in the differential,

$$\frac{1 \cdot 2 \cdot 3 \cdots (2n+1)}{(1 \cdot 2 \cdot 3 \cdots n)^2} \cdot \frac{1}{2^{2n} \sqrt{n}} \int_0^{2\delta\sqrt{n}\psi(r)} \left(1 - \frac{s^2}{n}\right)^n ds.$$

Let now $\delta\sqrt{n}$ be a finite quantity, $= \varrho$, so that the limit δ decreases in just the same measure as $\sqrt{n}$ increases. Then s remains finite within the assumed limit, however much n increases. For large n one may also put, by the development of logarithms in EULER's *Introductio*,

$$\left(1 - \frac{s^2}{n}\right)^n = e^{-s^2}$$

and

$$\frac{1 \cdot 2 \cdot 3 \cdots 2n}{(1 \cdot 2 \cdot 3 \cdots n)^2} = \frac{2^{2n}}{\sqrt{n\pi}},$$

according to EULER, *Calc. Diff.*, part II, chap. VI, §§160–162, as the limiting value to which it continually approaches as n increases. Hence the expression becomes

$$\frac{2n+1}{n\sqrt{\pi}} \int_0^{2\delta\sqrt{n}\psi(r)} e^{-s^2} ds,$$

for which one may safely write

$$\frac{2}{\sqrt{\pi}} \int_0^{2\delta\sqrt{n}\psi(r)} e^{-s^2} ds.$$

This is the expression for the probability that, in numerous observations arranged by magnitude, the middle error r lies between $r - \delta$ and $r + \delta$.

This probability will therefore be $1/2$, or the probable limits are given by

$$2\delta\sqrt{n}\psi(r) = \varrho,$$

that is,

$$\delta = \frac{\varrho}{2\sqrt{n}} \cdot \frac{1}{\psi(r)}.$$

For the law of errors assumed above,

$$\psi(\Delta) = 2\varphi(\Delta) = \frac{2h}{\sqrt{\pi}} e^{-h^2\Delta^2},$$

the probable limits of r are therefore

$$r \pm \frac{\varrho e^{h^2 r^2} \sqrt{\pi}}{4\sqrt{n} h}.$$

If, instead of $2n+1$, the number of observations is denoted by m , and if the equation

$$hr = \varrho$$

is used, this becomes

$$r \left\{ 1 \pm \frac{e^{\varrho^2} \sqrt{\pi}}{\sqrt{8m}} \right\}.$$

The numerical value of e^{ϱ^2} is 1.2554176, and the expression becomes

$$r \left\{ 1 \pm \frac{0.786716}{\sqrt{m}} \right\}.$$